Santiago del Palacio

Data Scientist & Astrophysicist

+46 070 013 7921
santiagodp1990@gmail.com
santimda.github.io
santiago-del-palacio
santimda
Gothenburg, Sweden

Professional Summary

Data Scientist & Scientific Software Developer with a background in astrophysics and 10+ years of experience in scientific computing, quantitative modelling, and Python development. Experienced in developing computational models, analysing complex datasets, and building reliable, reproducible software for demanding scientific problems.

Strong proficiency in Python, with experience in statistical modelling, machine learning, model evaluation, data pipelines, and automated testing. Developed an end-to-end breast cancer classification project comparing supervised learning models, evaluating sensitivity and specificity, optimizing classification thresholds, and packaging the selected model as a tested API using FastAPI and Docker.

Proven ability to lead technical projects, work independently and collaboratively, communicate complex technical concepts clearly, and learn new methods and technologies quickly.

Author of 9 first-author and 70+ co-authored peer-reviewed publications, including papers in Nature and Science Advances. Principal Investigator on 18 and Co-Investigator on 26+ competitively awarded international research projects. Delivered 18+ conference presentations internationally.

Seeking machine learning engineering roles where I can combine quantitative modelling, scientific software development, and machine learning to develop reliable computational solutions for challenging problems, particularly in medical technology and scientific applications.

Work Experience

04/2022 – 03/2026 **Postdoctoral Research Fellow**, *Chalmers University of Technology, Dept. of Space, Earth and Environment*, Gothenburg, Sweden

- Led an independent, cross-disciplinary research programme, defining scientific direction and coordinating collaborations across multiple institutions. This highly competitive fellowship also required working towards bridging multiple research groups within the division.
- Developed and validated physical models using statistical inference and numerical methods; quantified parameter uncertainties using Bayesian and frequentist approaches; developed tools for interpreting multi-wavelength observational datasets (radio, mm-wave, X-rays), with emphasis on reproducibility, version control, testing, and documentation.
- Co-supervised a Ph.D. student, guiding research design, data analysis, and problem-solving.
- Organised weekly departmental seminars to improve collaboration and knowledge sharing within the division.
- Served as referee for major journals and as member of international time-allocation committees for ALMA and *XMM-Newton*, evaluating project quality, technical feasibility and scientific impact, and providing constructive and actionable feedback.

04/2019 – 03/2022 **Postdoctoral Research Fellow (CONICET)**, *Argentine Institute of Radioastronomy*, Buenos Aires, Argentina

- Led an independent research programme, including project design, development of theoretical and numerical models, and coordinating international collaborations and multi-site observational campaigns.
- Designed and built an automated Python data-processing pipeline for long-term pulsar monitoring within the PuMA Collaboration.
- Supervised 3 M.Sc. students, defining project scope, work strategy, and continuous mentoring.

04/2014 – 03/2019 **Ph.D. Research Fellow (CONICET)**, *National University of La Plata*, La Plata, Argentina

- Developed numerical radiation-transport and non-thermal emission codes in FORTRAN for computational astrophysics, including code optimisation and OpenMP parallelization for computationally intensive simulations.
- Designed and executed multi-observatory campaigns; analysed and interpreted data from radio, X-ray, and gamma-ray facilities.
- Awarded the *Carlos M. Varsavsky Prize* for best Ph.D. thesis in astronomy in Argentina (2018–2019 cohort).

02/2011 – 03/2022 **Teaching Assistant / Head of Practical Assignments**, *National University of La Plata*, La Plata, Argentina

Taught several courses on Calculus, Algebra, and Astrophysics at undergraduate and graduate level for both science and engineering students. Student evaluations consistently remarked my clarity at explaining complex concepts and good predisposition overall.

Education

04/2014 – 11/2018 **Ph.D. in Astronomy**, *National University of La Plata*, La Plata, Argentina, Mark: 10/10

Thesis: *Non-thermal emission from systems harbouring massive stars*.
Award: *Carlos M. Varsavsky Prize* (best Ph.D. thesis in astronomy in Argentina, 2018–2019).

03/2008 – 03/2014 **Licenciatura in Astronomy (equiv. B.Sc. + M.Sc.)**, *National University of La Plata*, La Plata, Argentina, GPA: 9.56/10

Award: *Joaquín V. González Prize* (distinction for highest-achieving students, La Plata City Council).

Technical Skills

Programming	Python (advanced: NumPy, SciPy, scikit-learn, Pandas), Fortran (advanced), SQL, Bash
Machine Learning	Supervised learning, model evaluation, feature engineering, classification, neural networks/PyTorch
Software Engineering	Git/GitHub, Linux/Unix, automated testing, CI workflows, reproducible software, API development, Docker
Scientific Computing	Numerical modelling, statistical inference, uncertainty quantification, image analysis, signal processing, performance optimization

Languages

Spanish	Mother tongue
English	Fluent (C2)
Swedish	Upper Intermediate (B2/C1)

Outreach & Science Communication

2015 – 2026 14 public outreach activities, including public talks on astronomy and astrophysics (e.g. *Neutron Stars and Pulsars*, Slottsskogsobservatoriet, Gothenburg, 2026), lectures at the Swedish National Space School *Rymdforskarskolan* (2025), guided visits at open observatory nights, and co-authoring press releases.

References

Colleagues and line managers have highlighted my independence, contribution to a positive work environment, and capacity to move across different fields. References available upon request.